



CHEMISTRY

2AB 2013

Booklet 2

Name: _____

Multiple Choice answer sheet:

For each question shade the box to indicate the answer.

Use **only** a blue or black pen to **shade** the boxes.

For example, if b is your answer

1. ☐a ☒b ☐c ☐d

If you make a mistake, place a cross through that square, do not erase or use correction fluid.

Shade your new answer.

For example, if b is a mistake and d is your correct answer:

1. ☐a ☒b ☐c ☒d

In the event that you then change your mind back to your original answer, you then cross out the second selection and then circle the first choice.

For example, if b was the first choice and d your second, but you change your

mind back and b is your answer:

1. ☐a ☒b ☐c ☒d

1. ☐ a ☐ b ☐ c ☐ d
2. ☐ a ☐ b ☐ c ☐ d
3. ☐ a ☐ b ☐ c ☐ d
4. ☐ a ☐ b ☐ c ☐ d
5. ☐ a ☐ b ☐ c ☐ d
6. ☐ a ☐ b ☐ c ☐ d
7. ☐ a ☐ b ☐ c ☐ d
8. ☐ a ☐ b ☐ c ☐ d
9. ☐ a ☐ b ☐ c ☐ d
10. ☐ a ☐ b ☐ c ☐ d
11. ☐ a ☐ b ☐ c ☐ d
12. ☐ a ☐ b ☐ c ☐ d
13. ☐ a ☐ b ☐ c ☐ d

14. ☐ a ☐ b ☐ c ☐ d
15. ☐ a ☐ b ☐ c ☐ d
16. ☐ a ☐ b ☐ c ☐ d
17. ☐ a ☐ b ☐ c ☐ d
18. ☐ a ☐ b ☐ c ☐ d
19. ☐ a ☐ b ☐ c ☐ d
20. ☐ a ☐ b ☐ c ☐ d
21. ☐ a ☐ b ☐ c ☐ d
22. ☐ a ☐ b ☐ c ☐ d
23. ☐ a ☐ b ☐ c ☐ d
24. ☐ a ☐ b ☐ c ☐ d
25. ☐ a ☐ b ☐ c ☐ d

Section Two: Short answer**40% (80 marks)**

This section has 11 questions. Answer all questions. Write your answers in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time: 70 minutes.

Question 26**(5 marks)**

Four unlabelled powders were found in a laboratory and analysed to determine the identity of each. The four powders were known to be;



- (a) **Name (not formula)** the substance you would be able to identify by sight. **Explain**. (1)

Water was added to the remaining three unidentified solids. Only one did not dissolve.

- (b) **Name (not formula)** the substance that did not dissolve. (1)

Aqueous sodium carbonate solution was added to the remaining two unidentified solutions. A yellow solid was formed with one and a white solid was formed with the other.

- (c) **Name (not formula)** the white solid formed. (1)

- (d) Write the **ionic** equation for the reaction that produced the yellow precipitate. (2)

Question 27**(8 marks)**

Oxygen is one of the most abundant substances on Earth, and most elements have the ability to react with it. The table below shows two such oxides.

- a) Complete the table below for these species by drawing the structural formula, representing all valence shell electron pairs either as : or – . (2)

	Carbon dioxide (CO ₂)	Sulfur dioxide (SO ₂)
Structure (showing all valence electrons)		
Boiling point (°C)	- 57	- 10

- b) State one negative environmental impact of each of these gases. (2)

CO ₂	
SO ₂	

Another oxide found on Earth is silicon dioxide, SiO₂, which has a boiling point of 2230 °C.

- c) Why is the boiling point of this oxide so much higher than the two oxides shown in the table above? (4)

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Question 28

(7 marks)

The behaviour of two different acids, hydrochloric acid and phosphoric acid, were being compared and some of the results obtained are shown in the table below.

	0.5 mol L ⁻¹ HCl	0.5 mol L ⁻¹ H ₃ PO ₄
Rate of reaction with zinc metal	fast	Slow
Volume of acid required to neutralise 30 mL of 0.5 mol L ⁻¹ NaOH	30 mL	10 mL

- (a) Write the equation and observations for the reaction that would have taken place between the hydrochloric acid and the zinc metal. **Include in your answer only those species that react.** Describe in full what you would observe, where appropriate including colours, odours, precipitates and gases evolved. (3)

Equation: _____

Observations: _____

- (b) Explain why the zinc metal reacts more quickly with the hydrochloric acid. (2)

- (c) Explain why different volumes of each acid are required to neutralise the same amount of base. (2)

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Question 30**(7 marks)**

Blake bought a bottle of water to take on the plane with him. He drank it during the flight and put the lid back on. When the plane landed, he noticed that the bottle was a little crumpled and it looked as if the sides of the plastic bottle had been pushed inwards.

- (a) Explain, in terms of the kinetic theory, what has caused the bottle to become deformed like this. (3)

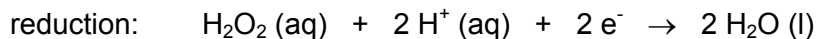
Airplanes often fly at an altitude similar to the peaks of very tall mountain ranges. At the top of Mount Everest, water boils at a temperature of 71 °C.

- (b) Explain why the boiling point of water is so much lower at this altitude. (2)

- (c) When a liquid is boiling, the temperature of the substance remains constant even though more heat is being added. Explain why this occurs. (2)

Question 31**(7 marks)**

The reaction between hydrogen peroxide and potassium bromide solution is a redox reaction that produces bromine, which can then be used to disinfect water supplies. The reduction half-equation for this process is shown below.



- (a) Write the oxidation half-equation for the reaction described above. (1)

- (b) Write the overall redox equation for this process by combining the oxidation and reduction half-equations. (1)

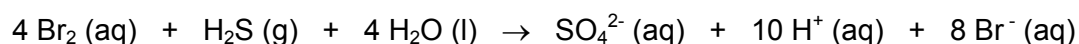
A sample of the bromine generated was added to a solution of sodium iodide.

- (c) Write the equation and observations for the reaction that would have taken place. Include in your answer only those species that react. Describe in full what you would observe, where appropriate including colours, odours, precipitates and gases evolved. (3)

Equation:

Observations:

Another reaction involving bromine is shown below.



- (d) Which substance is the reductant (reducing agent) in the reaction above? Support your answer with the use of oxidation numbers. (2)

Question 32**(6 marks)**

Calcium is the fifth most abundant element by mass in the Earth's crust. It is essential for all living organisms and is found in the body within bones and teeth. The table below shows some of the properties of calcium metal and the related compound calcium chloride.

	Ca	CaCl ₂
Melting point (°C)	842	772
Conductivity when solid	✓	✗
Conductivity when molten	✓	✓
Physical appearance (colour, lustre, texture etc)		

- (a) These two substances have similar melting points but quite different physical appearances. Complete the table above by adding a brief description of each substance. (2)
- (b) Draw the structural formula of calcium chloride, representing all valence shell electron pairs either as : or – . (1)

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- (c) Explain, in terms of structure and bonding, the difference in the ability of the two substances to conduct electricity. (3)

SEE NEXT PAGE

Question 33**(7 marks)**

Ozone (O_3) is an unstable allotrope of oxygen that was discovered in 1839 by Christian Shoenbein. It was named after the Greek word *ozein* meaning 'to smell' due to its sharp odour. When ozone gas reacts with potassium iodide solution and water this produces iodine, oxygen gas and aqueous potassium hydroxide.

- (a) Write a balanced full molecular equation for the reaction described above. (2)

- (b) If litmus indicator was added to the products of this reaction, what colour would you expect to see? (1)

One of the products, aqueous potassium hydroxide, is mixed with vinegar.

- (c) Write the ionic equation for the reaction that occurs. (1)

- (d) If potassium hydroxide is mixed with ammonium nitrate solution, describe in full what you would observe, where appropriate including colours, odours, precipitates and gases evolved. (1)

- (e) State and explain one precaution you should take when handling potassium hydroxide solution. (2)

Question 34**(7 marks)**

The three liquids in the table below have vastly different properties and uses.

	Ethanol (C ₂ H ₅ OH)	Pentane (C ₅ H ₁₂)	Water (H ₂ O)
Vapour pressure (kPa at 20 °C)	5.9	57.9	2.3

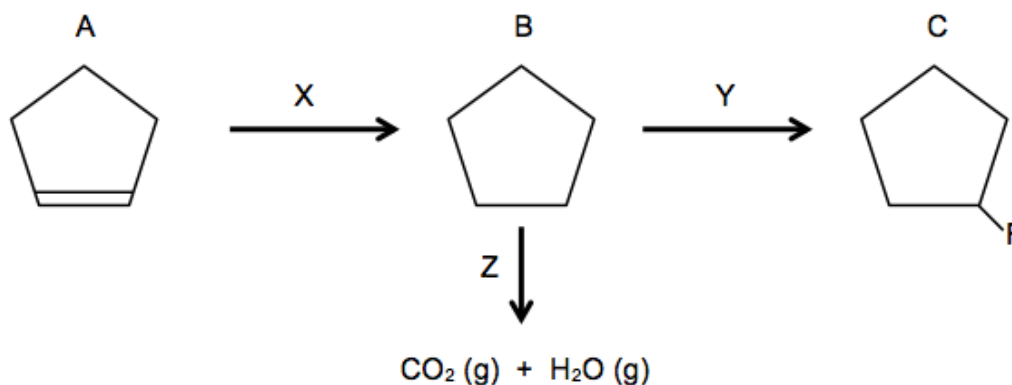
- (a) Explain, in terms of the kinetic theory, why it would be dangerous to keep a sealed canister of pentane in a non-refrigerated warehouse during summer. (3)

- (b) Which of the three substances in the table would have the highest boiling point? Explain your answer. (2)

- (c) When a little ethanol is mixed with water, it is observed that water no longer freezes at 0 °C. Explain why this is so. (2)

Question 35**(8 marks)**

Consider the diagram below;

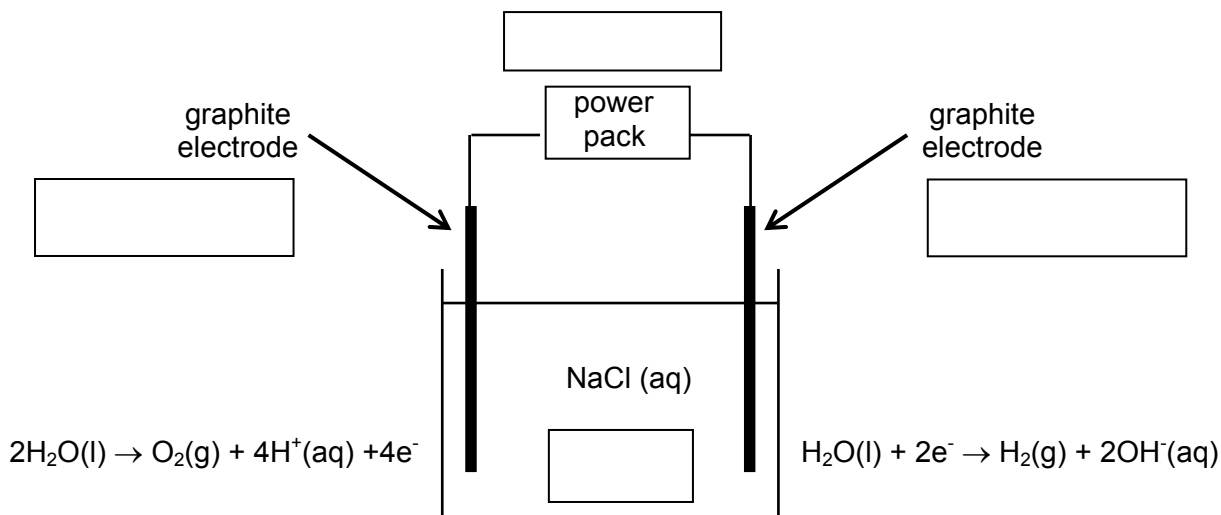


- (a) Name compound A. (1)
- _____
- (b) Name the reagent that has been added to allow reaction X to take place. (1)
- _____
- (c) Name the type of reaction occurring at X. (1)
- _____
- (d) Name the reagent that has been added to allow reaction Y to take place. (1)
- _____
- (e) Name the type of reaction occurring at Y. (1)
- _____
- (f) Name compound C. (1)
- _____
- (g) Write a balanced full molecular equation representing reaction Z. (1)
- _____
- (h) Under what conditions would carbon monoxide have been produced instead of carbon dioxide in reaction Z? (1)
- _____

Question 36

(8 marks)

An electrolytic cell was set up as shown in the diagram below, using aqueous sodium chloride as the electrolyte. **The reactions occurring at each electrode are also indicated in the diagram.**



- (a) On the diagram above label the **cathode** and **anode**, and label and **show the direction of electron flow** and the direction of anion flow. (3)
- (b) What is the function of the graphite electrodes? (1)

If **molten** sodium chloride, NaCl (l), had been used as the electrolyte instead of aqueous sodium chloride, NaCl (aq), very different reactions would have occurred at the cathode and anode.

- (c) Write the cathode and anode half equations for the molten NaCl cell. (2)

cathode: _____

anode: _____

- (d) Why are such different products formed in this case? (1)

- (e) What is the name of the industrial process where molten NaCl is electrolysed? (1)

End of Section Two

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Section Three: Extended answer**35% (70 marks)**

This section contains five (5) questions. You must answer all questions. Write your answers in the spaces provided below.

Where questions require an explanation and/or description, marks are awarded for the relevant chemical content and also for coherence and clarity of expression. Lists or dot points are unlikely to gain full marks.

When calculating numerical answers, show your working and reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

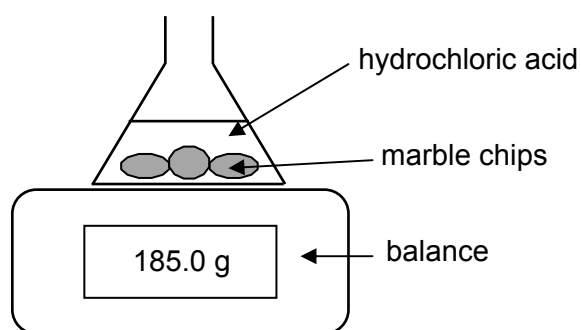
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Suggested working time: 65 minutes.

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Question 37**(13 marks)**

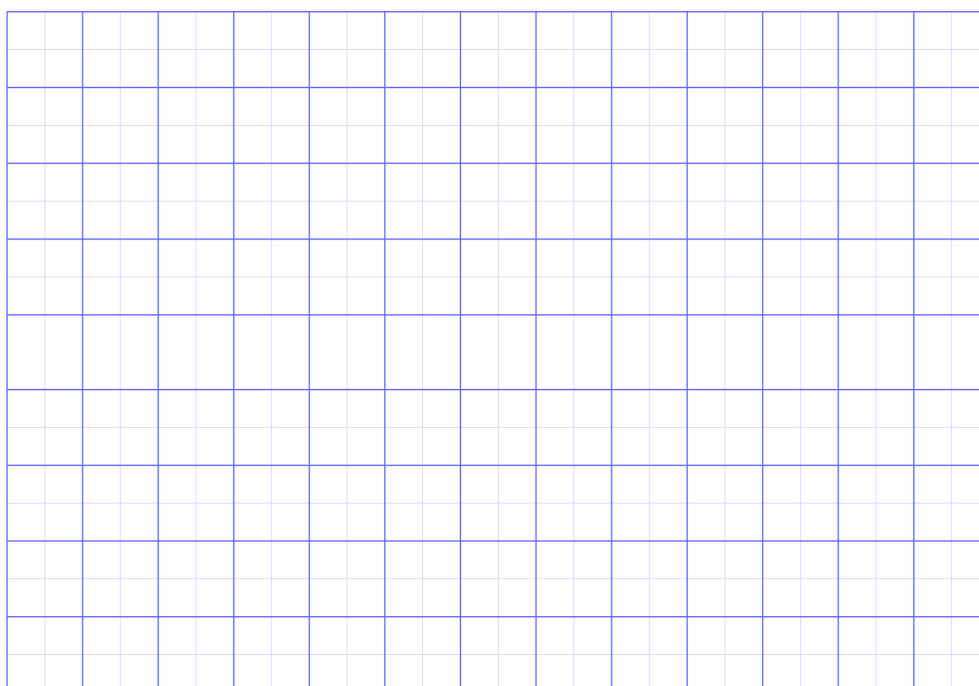
An experiment was designed in order to determine the rate of reaction between marble chips (mostly composed of calcium carbonate, CaCO_3) and hydrochloric acid. The equipment was set up as shown in the diagram below. 5.00 g of marble chips were placed into a conical flask containing excess hydrochloric acid and the mass of the flask was weighed at intervals until the reaction had completely stopped. The equation for the reaction that took place in the flask is also shown below. The results were recorded in the table.



Time (s)	Mass (g)
0	185
2	184.2
5	183.6
10	183.4
15	183.3
20	183.2
30	183.2



(a) Sketch a graph that illustrates the results of this experiment.

(4)**SEE NEXT PAGE**

- (b) Explain, in terms of the collision theory, why the reaction rate changes. How is this change in reaction rate related to the shape of the graph? (3)

- (c) What test could you perform to confirm that the gas produced was carbon dioxide? (1)

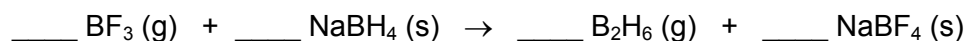
- (d) Calculate the mass of carbon dioxide produced from the marble chips. (1)

- (e) Calculate the percent by mass of calcium carbonate in the marble chips. (4)

Question 38

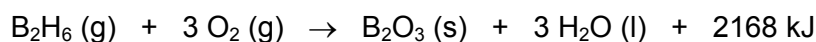
(15 marks)

Diborane, B_2H_6 , can be used as a rocket fuel due to the fact that it ignites easily with air. It is a colourless gas with a sickly sweet odour and can cause skin and eye irritation and shortness of breath if inhaled. There are many methods that can be used to prepare diborane and one of the most common is shown below. This reaction has a yield of about 30%.

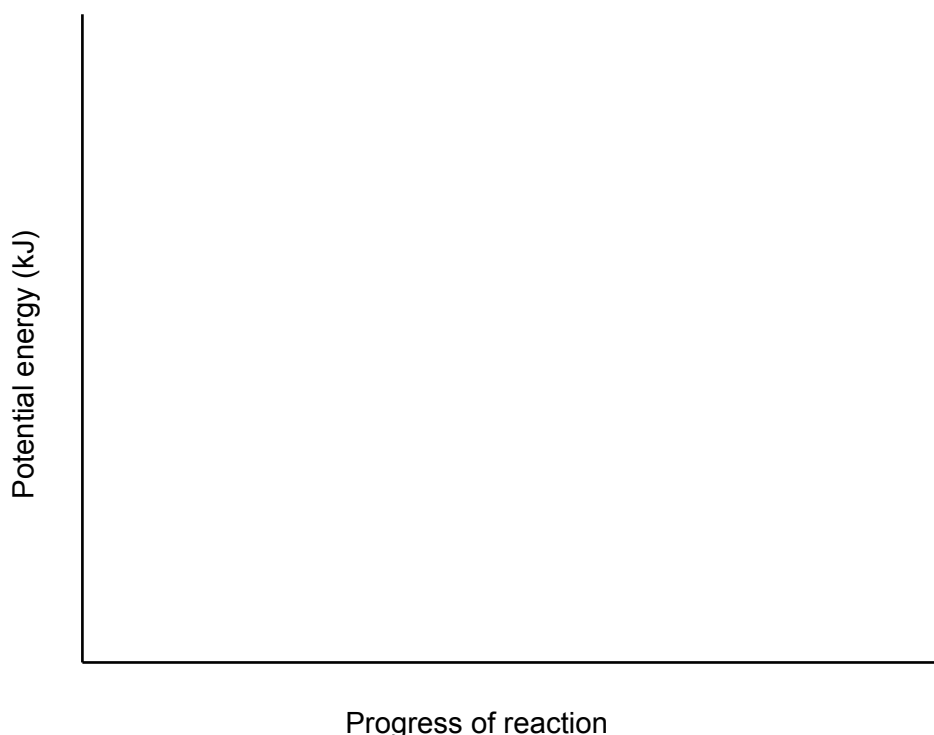


- (a) Balance the chemical equation above. (1)

The combustion equation for diborane, as would occur in the fuel tank of a rocket, is shown below. The activation energy for this reaction is 346 kJ.



- (b) Sketch a labelled potential energy diagram for the combustion of diborane. Label the activation energy, enthalpy change and transition state. (5)



- (c) State the value of the following; (2)

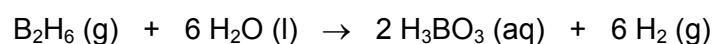
Activation energy (E_a) of the reverse reaction

Heat of reaction (ΔH) of the reverse reaction

SEE NEXT PAGE

- (d) Explain what information the heat of reaction (ΔH) for the combustion of diborane provides in terms of the bond-breaking and bond-making processes that occur. (3)

Diborane is also soluble in water and reacts to form boric acid and hydrogen gas according to the equation below.



- (e) If 9.27 L of diborane, stored at STP, was dissolved into 1.15 L of water, what concentration of boric acid would be produced? (Assume all the gas dissolves). (3)

- (f) What test could you perform to confirm that hydrogen gas was being produced? (1)

C₅H₁₀**Question 39****(13 marks)**

A hydrocarbon was sent for analysis to determine whether it was an alkane or alkene. The hydrocarbon sample was burnt in a stream of pure oxygen and this produced 0.321 g of carbon dioxide and 0.131 g of water vapour.

- (a) Calculate the empirical formula of the compound. (5)

- (b) If the molecular weight of the compound was known to be 42.1 g mol^{-1} , calculate the molecular formula of the compound. (2)

- (c) Draw and name the two possible isomers of the compound. (4)

Structure:	Structure:
IUPAC name:	IUPAC name:

- (d) When the compound was mixed with bromine water, $\text{Br}_2(\text{aq})$, no reaction was observed. Which of your 2 isomers must be the correct structure? Explain your answer. (2)

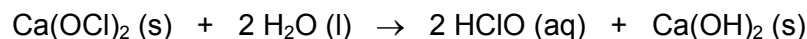
Question 40**(14 marks)**

Calcium hypochlorite, $\text{Ca}(\text{OCl})_2$, is a grey-white solid that can be used as a bleaching agent. It is often used to chlorinate pool water as it effectively kills bacteria and algae.

Raymond was trying to get his pool ready for summer, so he measured out 300 g of solid 'pool shock' and tipped it into his pool which had a capacity of 25 000 L. If the 'pool shock' was 65.0% calcium hypochlorite;

- (a) Calculate the final concentration of $\text{Ca}(\text{OCl})_2$, in mol L^{-1} , in Raymond's pool once all the 'pool shock' had dissolved. (3)

Once calcium hypochlorite is added to water it forms hypochlorous acid and calcium hydroxide as shown in the equation below.



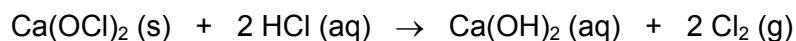
- (b) Hypochlorous acid is a monoprotic acid. Explain what this means, using an equation to support your answer. (2)

The calcium hydroxide formed is only slightly soluble in water and can clog up the pool filter, requiring the use of an anti-scaling agent to remove the solid.

- (c) Is calcium hydroxide a strong electrolyte, weak electrolyte or non-electrolyte? Explain your answer. (2)

SEE NEXT PAGE

Calcium hypochlorite can also be used to manufacture chlorine gas by heating with dilute hydrochloric acid, as shown in the equation below.



If 15.6 kg of calcium hypochlorite was heated with 98.0 L of 2.17 mol L⁻¹ hydrochloric acid;

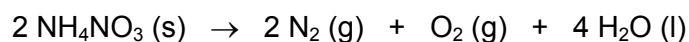
(d) Calculate the limiting reagent. (4)

(e) Calculate the mass of excess reagent. (3)

Question 41**(15 marks)**

Ammonium nitrate is manufactured in large quantities and has many different uses. It is the main ingredient in instant cold packs, it is commonly used in explosives as an oxidising agent, and due to the high nitrogen content it is also widely used in agriculture as a fertiliser.

Ammonium nitrate must be stored and transported carefully or it may decompose according to the reaction below. This process is very exothermic and can cause an explosion.



- (a) Why might the decomposition of ammonium nitrate cause an explosion (other than because it is an exothermic process)? (1)
-
- (b) If 825 g of ammonium nitrate was to completely decompose, calculate the **total volume of gas** that would be produced at STP. (3)

As stated above, ammonium nitrate can be used as a fertiliser as it is water-soluble and contains a high level of nitrogen, which is essential for plant growth.

- (c) Write an equation showing ammonium nitrate dissolving in water. (1)
-

- (d) Explain why a solution of ammonium nitrate is a good conductor of electricity whilst a solution of ammonia is not. (4)

A sample of ammonium nitrate solution is being diluted and prepared for use as a spray-on fertiliser in a vegetable garden.

- (e) What volume of 2.70 mol L^{-1} ammonium nitrate stock solution should be used to produce 9.00 L of a 0.155 mol L^{-1} working solution? (2)

- (f) If 5.00 L of the dilute 0.155 mol L^{-1} solution was sprayed evenly over a 2 m^2 vegetable garden, calculate the mass of nitrogen that could potentially be absorbed per cm^2 of soil. (Note: $1 \text{ m}^2 = 10\,000 \text{ cm}^2$) (4)

End of questions

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